Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Targeted Elimination of the Causative Agents of Body Malodor and Acne using Lytic Enzymes			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio-Medical		*Bio-Medical*	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	*Bio-Cosmetics / Materials*	*Medical Bio*	*Enzyme Pharmaceuticals*
Concept of Proposed Technology	Skin problems are often caused by single bacterial species. *Cutibacterium acnes* is the causative agent of acne, *Corynebacterium jeikeium* is responsible for body malodor, and *Staphylococcus aureus* (and its antibiotic-resistant strain, MRSA) cause non-healing skin lesions. Additionally, bacterial species like *Streptococcus mutans* cause bad breath and tooth decay, whereas *Streptococcus pyogenes* is the causative agent of strep throat. The current approach to managing these issues consists of regular use of broad-spectrum antibiotics, cleansers, disinfectants, and deodorants. These, however, have multiple adverse effects, such as disrupting the commensal skin microbiome, giving rise to resistant microbial strains, and skin damage. An emerging alternative approach to manage microbial contamination is the use of cell-wall lytic enzymes. These enzymes, from multiple classes, have evolved to recognize single species of bacteria. They kill the target bacteria by digesting its cell wall on contact – without affecting the integrity of the skin microbiome, and without giving rise to resistance. We propose to develop skin-			

	friendly cosmetic /cosmeceutical formulations, which will incorporate our lytic enzymes targeting specific problem bacterial species (*i.e.*, deodorant containing lytic enzymes with activity against malodor-causing *C. jeikeium*) (**Figure 1**). **Figure 1**. Proposed lytic enzymes-based applications.
Rationale for Joint R&D Project	We are leveraging our world-leading expertise in identifying and applying lytic enzymes. This includes enzymes with activity against various strains of Staphylococcus, Bacillus, Clostridia, Corynebacteria, Listeria, among others. We seek an industry partner to extend the use of lytic enzymes to specific commercial applications in various industrial biotechnology areas by advancing the proposed lytic enzyme-based platform for elimination of target bacterial species, with the goal of developing off-the-shelf consumer cleansing/cosmetic products. Specifically, the Korean SME would be expected to use their expertise and scale-up capabilities to incorporate highly selective lytic enzymes into cosmetic / cosmeceutical products such as skin lotions/cleansers/mouthwashes. Currently, the basic research has been completed and initial scale-up of enzyme productions has been performed. Moreover, applications described above have been developed at the lab scale.
Key Words	1. Anti-microbial lytic enzymes 2. Topical skin treatment 3. Antibiotic resistance 4. Cosmeceuticals 5. Recombinant proteins

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

We propose to develop commercial applications based on our modular cell-lytic enzyme platform as active antimicrobial agent. Our group has extensive experience identifying, producing, and testing lytic enzymes which target specific bacterial species. We have discovered lytic enzymes targeting the environmental pathogen *Bacillus anthracis* (Mehta et al., 2013), the intestinal bacterium *Clostridium difficile* (Mehta et al., 2016), demonstrating activity even against antibiotic-resistant isolates; the body malodor-causing *Corynebacterium jeikeium* (Mixon et al., 2025), and *Mycobacterium tuberculosis*, the causative agent of tuberculosis (Grover et al., 2014), *etc*. We have also tested our lytic enzymes in a range of applications, from anti-MRSA skin bandages (Miao et al., 2011, **Figure 2)**, to Listericidal food product packaging (Solanki et al., 2013), to active nanoscale composites for surface decontamination (Pangule et al., 2010).

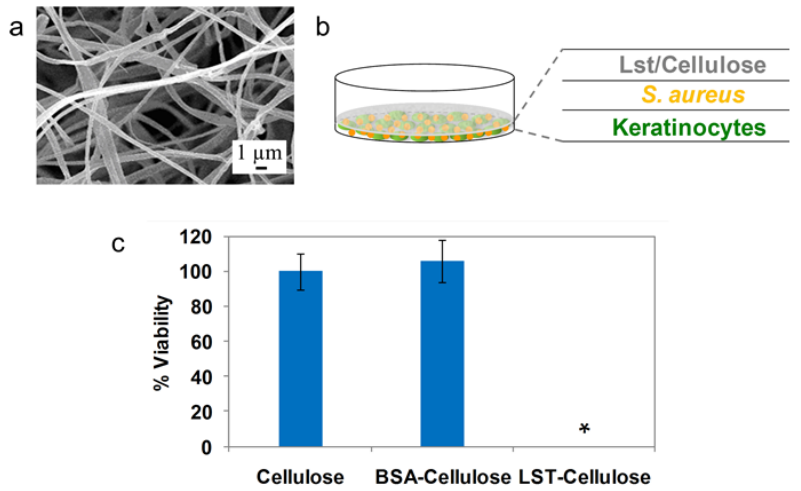

Figure 2. (a) SEM image of oxidized cellulose fibers. (b) A skin model to assess the anti-bacterial activity of Lst-functionalized cellulose mats - a three-layer model consisting of keratinocytes, S. aureus suspension, and cellulose mats. (c) % Viable *S. aureus* cells present on different cellulose-based mats. (*) no colonies were observed in these samples, suggesting complete sterilization.

The overarching goal of the current work is to use lytic enzymes targeting specific bacterial species in relevant consumer products, as follows:

Targeted bacteria	Causative agent of:	Proposed application
Cutibacterium acnes	acne	Facial creams/cleansing lotions
Corynebacterium jeikeium	body malodor	Deodorants/underarm protection
Staphylococcus aureus (MRSA)	skin infection/lesions	Skin bandages/wound-healing
Streptococcus pyogenes	strep throat	Mouthwash
S. mutans	bad breath/tooth cavities	Mouthwash

R&D Target Specifications	1. **Production of lytic enzyme:** • The target is development of a scalable fermentation process for production of recombinant enzyme in microbial expression system. 2. **Matrix formulation:** • Incorporate active component (lytic enzyme) into desired matrix (e.g., water, glycerin, PEG, palmitate, EDTA), such as the lytic enzyme retains catalytic activity, as determined by standard *in vitro* assay. The matrix composition should be iteratively tailored to the optimal activity conditions of the lytic enzyme. 3. **Validation of functional parameters:** • The active formulation should demonstrate ≥ 99.9% activity against target bacterial species at a cell challenge of 10^6 cells/mL; hence reduce to no more than 10^3 cels/mL. • The active formulation should demonstrate ≥ 95% specificity against target bacterial species. 4. **Safety and biocompatibility:** • The active formulation should not have adverse effects higher than that of the matrix alone. 5. **Operational requirements:** • The active formulation should have shelf life of at least 180 days at room temperature. • The formulation should retain activity at ambient and physiological temperature, 21–37 °C.
R&D Plan	The objective of this R&D effort is to advance commercial applications based on our modular cell-lytic enzyme platform as active antimicrobial agent. We start at TRL 3 (lab scale proof-of-concept) to TRL 5 (component and system validation in a relevant environment). The R&D plan is organized into five technical tasks, with an initial phase covering a 6-month period. **Task 1. Identification and initial assessment of potential applications (Months 1-6)** The aim of this Task is to identify one or more target applications to incorporate lytic enzymes as an active ingredient. These applications can be cosmetic (*e.g.*, facial lotion, deodorant), cosmeceutical (*e.g.*, wound-healing ointments, bandages), household (*e.g.*, surface decontamination product), or industrial (e.g., self-decontaminating food packaging). The selected application will determine the composition of the matrix in which the active component (lytic enzyme) will be incorporated: • The lytic enzyme can be incorporated as a soluble component of an aqueous solution, or • It can be encapsulated as a dry formulation within the matrix. **Deliverable**: A full proposal with a focus on one or more of the specific application areas indicated above. This part of the project will be completed jointly by RPI and a Korean SME. **Milestone**: An assessment of the chosen application area(s) in terms of likely use needs. **Task 2. Selection of lytic enzyme(s) (Months 7-12)** Based on the desired application, we will identify the optimal lytic enzyme (or cocktail of lytic enzymes) out of the genetic library of enzymes identified and tested in our lab, as well as additional candidates using bioinformatic analyses. Perform initial production of enzymes from an *E. coli* expression system currently used in the Dordick Lab. **Deliverable**: A set of at least two lytic enzymes identified and produced at bench scale for each

application chosen in Task 1. This part of the project will be completed by RPI in consultation with the Korean SME.

Milestone: Each lytic enzyme prepared at the bench scale with functional parameters listed in R&D targets.

Task 3. Production of lytic enzyme(s) for R&D purposes (Months 13-25)

Task 3a. For R&D purposes, we will produce the selected lytic enzyme in *E. coli* expression system. The Origami B (DE3) Tet^R, Kan^R, strain is ideal for being transformed with vector pH-MBP-3c (Amp^R), carrying the lytic enzyme gene under the control of inducible *tac* promoter. We will perform fed-batch fermentation in a 3-L Eppendorf BioFlo 320 reactor, to produce the protein of interest. We will purify it using affinity chromatography, perform QC (determine concentration, quantify lytic activity), and will freeze standard aliquots for the duration of the work. These aliquots will be used in all subsequent R&D work.

Task 3b. Origami B (DE3) strain is a derivative of *E. coli* K-12, which is extensively used as a microbial factory to produce complex recombinant proteins like insulin, growth hormones, and vaccines. However, the expression system requires three antibiotics for selection. Therefore, after the selection of optimal lytic enzyme, we will change the expression system to one using essential gene complementation (rather than antibiotics) to exert selective pressure.

Task 3c. Preparation for pilot-scale production of lytic enzyme. The procedure for fed-batch microbial fermentation will be optimized for the specific lytic enzymes to be produced. We have bioreactors ranging from 1-30 L in size, which will allow us to scale up the process, with a view of linearly scale up to industrial-volume fermentations. We will develop a Standard Operating Procedure (SOP) for technology transfer.

Deliverables:

1. Purified lytic enzyme preparation, with quality control (QC) parameters matching the R&D specifications, to be used during this collaboration,
2. Enzyme-producing strain, compliant with industry standards,
3. Scale-up parameters for production of antimicrobial lytic enzyme.

This work will be done at RPI in consultation with the Korean SME.

Milestone: The production of each lytic enzyme prepared at the bench scale to be scaled up to a fed-batch fermentation process.

Task 4. Iterative optimization of the matrix composition in the context of lytic enzyme activity (Months 18-25)

A typical cosmetic formulation consists of 10 or more individual chemical compounds. Any of them may adversely affect the lytic activity of the active component (enzyme). We will undertake iterative optimization – likely in a matrix format – where commercially acceptable formulations will be used as a matrix to suspend the lytic enzyme, and test its lytic activity *in vitro*. The optimization steps, followed by *in vitro* testing, will be repeated until optimal formulation is identified. We anticipate the need of ongoing interaction with the industry partner at this step, in order to identify an acceptable compromise between optimal product formulation and optimal enzyme activity.

Deliverable: Product formulation with standardized functional parameters. The formulation

	development will be performed by the Korean SME, with testing of target formulations performed by RPI. **Milestone**: The selection and *in vitro* testing of the product formulation components in presence of the antimicrobial enzyme will ensure optimal operational activity. **Task 5. Incorporation of lytic enzyme(s) into specific matrix and operational testing (Months 26-36)** **Task 5a.** The lytic enzymes are proteins, and, as such, they might induce immune response. This is an undesirable side effect. It is typically evaluated using small animal model. We will perform *in vitro* immunogenicity studies using a simplified mammalian cell culture system, as used in preclinical testing of protein therapeutics (Walsh et al., 2024). **Task 5b.** The operational evaluation will include stability testing of the formulation under a defined range of conditions (to be determined between the partners in this collaboration), and testing of the retention of functionality (antimicrobial activity) **Deliverable**: Fully tested product formulation(s). The formulation, including the active ingredient(s), will be prepared by the Korean SME, the testing and characterization will be performed at RPI. **Milestone**: Operational characterization of the product formulation.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Materials:**
	• Identify and produce a set of lytic enzymes identified, expressed, and tested in Dordick lab. Identify optimal enzyme for target application:
	  ○ This part of the work will be performed based on a continual feedback loop between the SME and the academic lab.
	• Develop microbial expression system to produce lytic enzyme(s):
	  ○ This work will be informed by the requirements of the specific consumer product, and will be done in consultation with the SME partner.
	• Produce, purify, and QC sufficient amount of enzyme (1-2 g) for the R&D stage of work.
	
	Methods:
	• Activity testing and data collection: perform iterative testing to optimize activity of the enzyme formulation:
	  ○ The assays to determine the antimicrobial enzyme activity as a constituent of a complex formulation will be performed at the academic lab, in a feedback loop with the SME partner.
	• Biological testing: perform *in vitro* testing using eukaryotic cell culture system to determine the presence of any adverse effects (*i.e.*, immunogenicity):
	  ○ This work can be performed at the academic lab. Should the SME have the relevant expertise, it can be shared between the partners.
	  ○ Alternatively, should the SME partner have the relevant expertise, they can perform validation of the results obtained by the academic lab, to ensure reproducibility.

	Team: • **Jonathan S. Dordick, Ph.D.,** is Institute Professor of Chemical and Biological Engineering at RPI, with a joint appointment in the Department of Biological Sciences. He also serves as Vice President for Strategic Alliances and Translation. He received his B.A. degree in Biochemistry and Chemistry from Brandeis University and his Ph.D. in Biochemical Engineering from the Massachusetts Institute of Technology. At RPI, he has served as the Vice President for Research, founding Co-Director of the Rensselaer-Mount Sinai Center for Engineering and Precision Medicine, Director of the Center for Biotechnology & Interdisciplinary Studies, and Department Chair. Dr. Dordick has made foundational contributions to protein science and enzyme technology, biomanufacturing of therapeutic proteins and nucleic acids, and drug discovery. Dr. Dordick has received numerous awards and honors, including election to the National Academy of Engineering and National Academy of Inventors, and receipt of numerous society awards. He has founded several companies and published over 460 papers and is an inventor/co-inventor on over 50 patents. • **Elena Paskaleva, Ph.D.,** is a Senior Research Scientist at the Center for Biotechnology and Interdisciplinary Studies at RPI. She received her M.S. degree in Cellular and Molecular Engineering from the University of Sofia, Bulgaria, and her Ph.D. in Pharmacogenomics from the University of Manchester in England. Dr. Paskaleva's expertise is in the field of therapeutic proteins and drug discovery. She has over a decade of experience in identifying antimicrobial lytic enzymes, which have been developed and tested against a range of bacterial pathogens such as *B. anthracis*, *C. difficile*, and MRSA. She also has significant experience in the development of antiviral compounds of natural origin, and with publications on HIV-1 inhibition. At RPI, Dr. Paskaleva has served as a co-director of the clinical COVID-19 testing lab and was a foundational member of its Applied Research Center. • **Marc Douaisi, Ph.D.,** is Senior Research Scientist at RPI in the Center for Biotechnologies and Interdisciplinary Studies (CBIS). He received his B.Sc. degree in Cell & Molecular Biology from the Nantes University (France) and his M. Sc. and Ph.D. degrees in Immunology from the Aix-Marseille University (France). Prior to joining RPI, his research focused on HIV infection and human T cell development in the thymus, especially regulatory $CD4^+$ T cells. As a member of the Applied Research Center at RPI, Marc Douaisi focused on protein biology and bio-enzymatic processes for the development of anti-microbial biomaterials, and the bio-manufacturing of synthetic heparin. In 2020, he established the RPI's COVID-19 testing lab, and served as its co-director.

Expected R&R of Korean SME Partner	We are looking for an industry partner with expertise in development of consumer product formulations such as cosmetics, cosmeceuticals, skin care products. An ideal partner will have:
	• Ample experience in formulating skin care, cosmetics, or oral health products:
	  ○ This experience will be relevant for the translation of biologically active molecule (antimicrobial lytic enzyme) into a consumer product.
	  ○ The academic lab will rely on the experience of the SME in formulating cosmetic/ skin-case/ cosmeceutical consumer products, and their expertise in combinatorial selection of optimal compounds.

	• *In vivo* testing platform for biocompatibility of the product:
	  ○ While not all cosmetic/ skin care products require animal testing, it is prudent to evaluate the active ingredient, as well as the overall formulation, for any adverse effects it might have, which are not readily detectable with *in vitro* /*ex vivo* testing assay.
	• Microbial fermentation/protein purification capacity:
	  ○ The active compound (antimicrobial lytic enzyme) is produced as a recombinant protein in a fed-batch fermentation. At an industrial scale this work can be performed either by the Korean SME partner, or it can be outsourced to a subcontractor.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Breakthroughs in Management of Skin Conditions and Infections** All currently available methods for skin cleansing, wound management, oral hygiene, and household surface decontamination are based on chemical agents with broad antimicrobial activity (i.e., they are alcohol-based, acid- or alkali-based, or contain conventional antibiotics). This has resulted in reduction of the beneficial skin microflora, and gained resistance to conventional antibiotics, which gradually reduces the effectiveness of this broad-spectrum approach. In contrast, the use of cell wall lytic enzymes for targeted elimination of problematic microbial species has the following advantages: 1. Gained resistance against lytic enzymes is unlikely compared to that against conventional antibiotics. 2. Lytic enzymes are highly specific, and therefore, unlikely to have adverse activity against commensal microflora. Moreover, lytic enzymes cannot target human cells, since the mammalian peptidoglycan is structurally different from the microbial peptidoglycan. Hence, we do not expect enzymatic action to result in adverse impacts on human health. However, we will evaluate the toxicity and optimize the biocompatibility in the proposed work. 3. The lytic enzymes are active against both proliferating and dormant (post-antibiotic) bacterial cells. **Commercialization** From a commercial perspective, this collaboration will help to develop a new range of cosmetic/cosmeceutical products for personal care, and wound healing. While enzyme-based cosmetics are increasingly common, using lytic enzymes to target specific constituents of the skin microbiome – while maintaining its integrity – is a paradigm-changing advance in skin care and personal hygiene. While the lytic enzymes we have identified and tested are of natural origin, any further genetic modifications we design (such as reducing immunogenicity, increasing solubility), will serve as a basis to generate intellectual property. Significantly, unlike many therapeutic proteins, the lytic enzymes are readily produced by microbial fermentation. This results in several orders of magnitude lower production costs compared to proteins

	produced in a typical industrial eukaryotic expression (e.g., in CHO cells). **Broader Industrial and Societal Impact** On a broader scale, antibiotic resistance is rapidly becoming a global problem. Even minor cuts and skin wounds are treated with ointments such as Neosporin or Fucidin, with broad-spectrum activity, leading to gained antibiotic resistance. By developing skin care products with active ingredients that can eliminate only the target constituent of the microbial milieu, this collaboration can help mitigate the further spread of antibiotic resistance, while developing commercial products that can be continuously used without diminishing their efficacy, thus creating significant social value and improving global public health outcomes.

6. References

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Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization